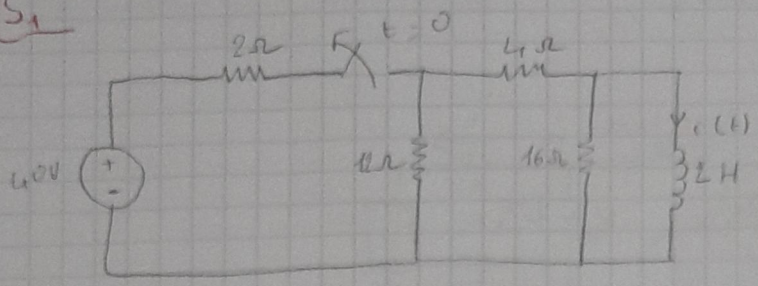
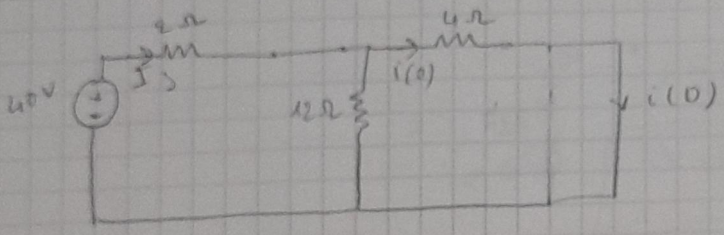
S1



$i(t) = ?$

$t < 0$ için



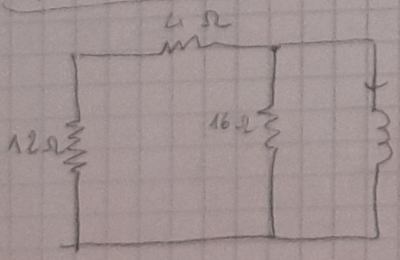
$R_{eq} = 5\,\Omega$

$I_s = \frac{40}{5} = 8A$

$i(0) = I_s \cdot \frac{12}{16} = 6A$ (Akım bölmeden)

$\boxed{i(0) = 6A}$

$t \geq 0$ için



$z = \frac{L}{R_{th}} = \frac{1}{4}$

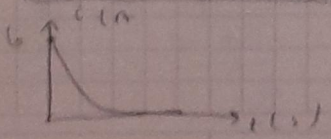
$(12+4) // 16 = 8 = R_{th}$

$i(\infty) = 0$

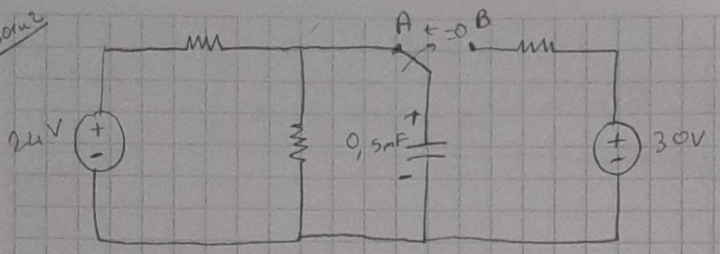
$x(t) = x(\infty) + [x(0) - x(\infty)]\, e^{-t/z}$

$i(t) = 0 + [6 - 0]\, e^{-4t}$

$\boxed{i(t) = 6e^{-4t}\ A}$

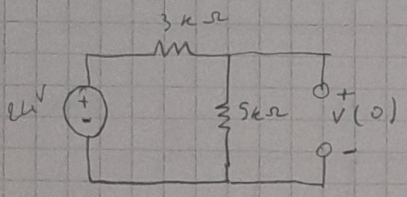


Soru 2



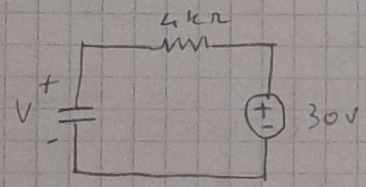
$v(t) = ?$ $v(1) = ?$ $v(4) = ?$

$t<0$ için



$$V(0) = \frac{5}{8} \cdot 24 = 15\,V$$

$t>0$ için



$$V(\infty) = 30\,V$$

$\tau = R_{th} \cdot C$

$R_{th} = 4k\Omega$

$\tau = 4 \cdot 10^3 \cdot \frac{1}{2} \cdot 10^{-3}$

$\tau = 2s$

$$v(t) = 30 + [15 - 30]\, e^{-t/2}\ V$$

$$= 30 - 15\, e^{-t/2}\ V$$

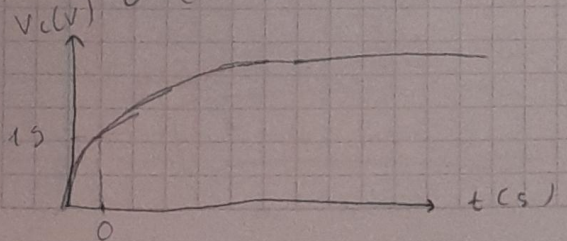
$v(1) = 20,9\,V$

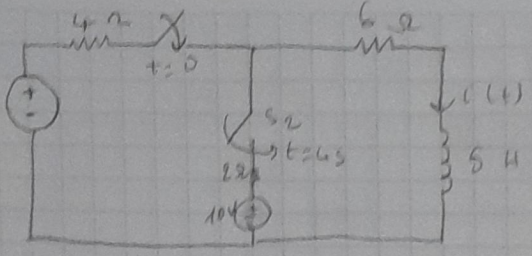
$v(4) = 27,97\,V$

$v(0) = 15\,V$

$v(10) = 29,89$

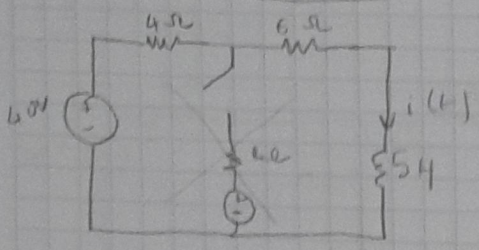
$v(100) = 29,999$





$t<0$ için $i_L(0) = 0A$

$0 \le t < 4$ devre



$i = \frac{40}{10} = 4A$

$R_{th} = 10\Omega$

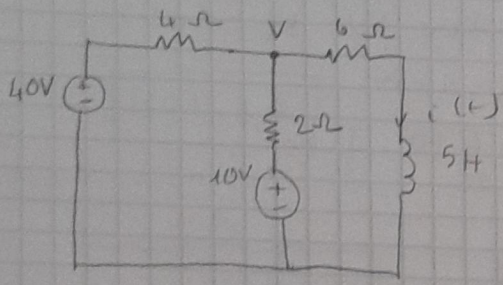
$\tau = \frac{L}{R_{th}} = \frac{5}{10} = \frac{1}{2}s$

$i(t) = 4 + [0 - 4]e^{-2t}$ A

$= 4(1 - e^{-2t})$ A , $0 \le t < 4$

$t \ge 4$ için devre

$i(4) = 4A$ (ikinci aralık için başlangıç koşulu)



$\frac{40 - V}{4} + \frac{10 - V}{2} = \frac{V}{6}$

$V = \frac{150}{11}$ V

$i(\infty) = \frac{\frac{150}{11}}{6} = 2,72$ A

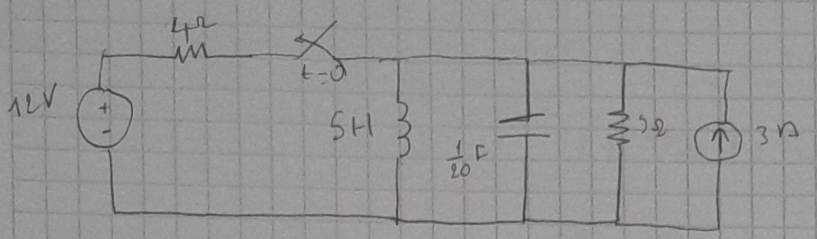
$R_{th} = \frac{22}{5}\Omega$ $\tau = \frac{5}{22}s$

$x(t) = x(\infty) + [x(0) - x(\infty)]e^{-(t-t_0)/\tau}$

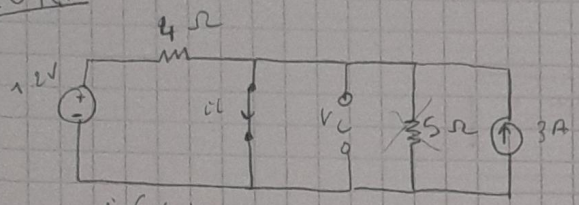
$i(t) = 2,72 + [4 - 2,72]e^{-(t-4)/(\frac{5}{22})}$ A $t \ge 4$ için

$$i(t) = \begin{cases} 0, & t < 0 \text{ için} \\ 4(1-e^{-t}), & 0 < t < 4 \text{ için} \\ 2,72 + 1,27\, e^{-(t-4T/(T=2))}, & t > 4 \text{ için} \end{cases}$$

Örnek



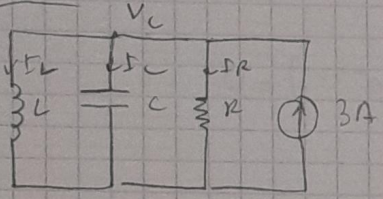
t<0 için



$i_L(0) = 6A$

$V_L(0) = 0V$

t>0 için



$V_L = V_L$

$V_L = L\frac{di_L}{dt}$ $\quad i_C = C\frac{dV_L}{dt}$

$I_L + I_C + I_R = 3$

$i_L + C\frac{dV_L}{dt} + \frac{V_L}{R} = 3$

$$\boxed{\frac{i_L}{C} + \frac{d\,V_L}{dt} + \frac{V_L}{RC} = \frac{3}{C}}$$

$$\Rightarrow \frac{i_L}{C} + L\cdot\frac{d^2 i_L}{dt^2} + \frac{L}{RC}\frac{di_L}{dt} = \frac{3}{C}$$

$$\Rightarrow \frac{i_L}{CL} + \frac{d^2 i_L}{dt^2} + \frac{1}{RC}\frac{di_L}{dt} = \frac{3}{LC}$$

$$\frac{d^2 i_L}{dt^2} + \frac{1}{RC}\cdot\frac{di_L}{dt} + \frac{i_L}{CL} = \frac{3}{LC}$$

$$\frac{d^2 i_L}{dt^2} + 4\frac{di_L}{dt} + 4 i_L = 12$$

$$\boxed{s^2 + 4s + 4 = 0}$$

$s_1 = s_2 = -2$

Soit $s = -2$

$i_{Ln}(t) = (A_1 + A_2 t) e^{-2t}$

$i_L(t) = i_{Ln}(t) + \frac{i_{Lf}(t)}{3}$

$i_{Lf}(t) = B$

dérivée de $y = ax^2 \Rightarrow y' = 2ax$

$0 + 0 + 4B = 12$

$B = 3$

$i_L(t) = (A_1 + A_2 t) e^{-2t} + 3$

$i_L(0) = 6 \Rightarrow 6 = A_1 + 3$

$A_1 = 3$

$$\boxed{i_L(t) = (3 + A_2 t) e^{-2t} + 3}$$

$V(0) = 0$

$V_L = L \frac{di_L}{dt}$

$\Rightarrow 5\left(A e^{-2t} - 2(3 + A_2 t) e^{-2t}\right) = 0$

$\Rightarrow 5(A_2 - 6) = 0$

$A_2 = 6$

$$\boxed{i_L(t) = (3 + 6t) e^{-2t} + 3}$$